

Nifasath Ahamed

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Experience

Agilent Technologies

Wilmington, DE

R&D Engineer

Jan 2023 – Present

- Designed and deployed **production-grade Flask REST APIs** processing **10K+ hardware alerts/day**, improving system throughput by ~40% and enabling scalable, distributed alert processing
- **Led architecture and implementation of a Retrieval-Augmented Generation (RAG) system**, enabling sub-second document retrieval across internal knowledge bases and reducing researcher lookup time by ~50%
- Built **NLP-driven document retrieval pipelines**, improving search relevance and reducing manual troubleshooting time by ~30%
- Developed **automated Linux-based memory diagnostics system**, reducing root-cause analysis time from hours to minutes (~60% improvement)
- Designed and implemented **distributed system capabilities** (remote reboot, firmware updates, real-time monitoring), reducing on-site interventions by ~35% across lab environments for custom hardware systems

Cerebras Systems

Sunnyvale, CA

Member of Technical Staff

March 2021 – August 2022

- Built **CI/CD pipelines using Jenkins**, reducing manual deployment effort by ~50% and improving release reliability
- Developed **containerized distributed execution framework (Docker + AWS EC2)**, parallelizing workloads and reducing experiment turnaround time by ~40%
- Engineered **Linux-based performance monitoring tools**, enabling visibility into system bottlenecks and improving system efficiency by ~25%
- Diagnosed and resolved **system-level performance issues**, reducing runtime inefficiencies and improving compute utilization across ML workloads

IBM

Bangalore, India

Associate System Engineer

September 2018 – May 2019

- Developed data **pipelines using Kafka**, processing **50GB+ of streaming data daily** for enterprise applications
- Optimized **SQL queries and data models**, reducing query response times by ~30% and improving overall data pipeline performance

Skills & Interests

Languages and Backend: Python (primary), C, Javascript, Flask

Backend & APIs: Flask, REST API design

Cloud & DevOps: AWS (EC2/R2), Docker, Kubernetes, Jenkins, Azure AI Foundry

Database & Messaging: MySQL, MongoDB, Redis, Kafka

AI/ML: PyTorch, RAG pipelines, Model Quantization, Model Finetuning

Tools and Technologies: Jenkins, Docker, GDB, Jira, Git, Visual Studio

Databases: MySQL, Redis, MongoDB, SQLite

Machine Learning: PyTorch, TensorFlow, LangGraph, numpy, pandas

Education

Carnegie Mellon University

Pittsburgh, PA

Masters in Software Engineering – Scalable Systems

December 2020

Capstone: Adaptive Reinforcement Learning Agent with Lockheed Martin (Patent Pending)

Anna University

Chennai, India

Bachelors of Technology in Information Technology (Honor Roll)

May 2018